Summary

The immune system protects the host from infectious agents. Entry, survival and proliferation of pathogenic microbes are referred to as infections. Bacterial, viral and fungal pathogens predominantly cause infectious diseases. Recovery from these infections is naturally achieved with the help of the natural defence system of the host , that is the immune system. The ability of the immune system to protect the host from infections is known as immunity.

There are two major types of immunity, namely innate and acquired immunity. Innate immunity is a type of immunity obtained by birth, but it is non-spectific in its action. Innate immunity is provided by four barrier systems namely anatomical, physiological, phagocytic and inflammatory barriers.

Acquired immunity is a specific type of immunity which is provided by immune cells like T-Cells, B-Cells, K-Cells, NK-Cells etc., and immune components like antibodies. Acquired immunity is further divided into humoral immunity and cell mediated immunity. Antibodies play a vital role in humoral immunity. Immune cells like T-Cells play a vital role in cell mediated immunity.

Antigens are foreign substances which have an ability to induce production of immune response products.Depending upon their nature and origin, there are about thirteen different types of antigens. Antigenic determinant or epitope is the region of the antigen recognized by antibodies or T-Cell Receptor (TCR) of T cells. The ability of an antigen to induce an immune response is known as antigenicity. The factors that influence antigenicity are mainly of two types i.e. factors contributed by antigens and factors contributed by host cells.

Antibodies are proteins produced by B cells in response to a specific antigen. The IgG molecule is composed of two identical heavy (H) chains and two identical light (L) chains. There are about five isotypes of antibodies namely Ig G, Ig M, Ig A, Ig E and Ig D. The H and L chains are linked by disulfide bridges. Antigen-Antibody reactions refer to the interaction between antigens and antibodies.

The interaction between an antibody and a soluble antigen in aqueous solution forms a lattice that eventually develops a visible precipitation. Antibody that forms precipitation is known as precipitin. The interaction between antibody and a particulate antigen results in visible clumping called agglutination**.** Antibodies that produce such reactions are called agglutinins**.**

A blood type is a classification of blood, based on the presence and absence of antibodies in blood and inherited antigenic substances on the surface of red blood cells. Karl Landsteiner discovered ABO blood group system. The membranes of human red cells contain a variety of blood group antigens**,** which are also called agglutinogens. The most important and best known of these are the Aand B antigens.

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